HRMS

06 專案管理 — 系統設計書

HR06 PROJECT Service — System Design

系統名稱	版本
人力資源暨專案管理系統	**1.0**
日期	作者
2026-03-17	**HRMS 開發團隊**

目錄

專案管理服務系統設計書

版本: 1.0
日期: 2025-12-07
Domain代號: 06 (PRJ)
導入階段: 第二階段（專案管理核心） **目標:** 提供工程師完整的系統實作規格，供PM建立工項清單

目錄

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9. API設計
10. 工項清單摘要

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1. 服務概述

1.1 服務定位

專案管理服務負責客戶管理、專案建立與維護、多層級工項管理(WBS)及專案成本追蹤。這是軟體公司專案成本核算的關鍵服務，需與工時服務緊密整合。

1.2 核心功能

- ✅ **客戶資料管理:** 客戶基本資料、聯絡人管理
- ✅ **專案管理:** 類型、預算、時程管理
- ✅ **WBS工項管理:** 支援5層級工項結構
- ✅ **專案成員管理:** 角色指派、工時分配
- ✅ **專案狀態追蹤:** 進度、成本監控

- ✅ **專案成本分析:** 與工時整合計算人力成本

1.3 技術架構

- **前端:** ReactJS + Redux + Ant Design
- **後端:** Spring Boot 3.1.x + MyBatis
- **資料庫:** PostgreSQL 15.x
- **事件匯流排:** Kafka

1.4 服務邊界

屬於本服務	不屬於本服務
客戶資料管理	工時回報 (Timesheet Service)
專案資料管理	成本計算細節 (需整合工時+薪資)
工項(WBS)定義	
成員指派	
進度追蹤	

1.5 專案類型與預算模式

專案類型	說明
DEVELOPMENT	新開發專案
MAINTENANCE	維護專案
CONSULTING	顧問專案

預算模式	說明
FIXED_PRICE	固定價格，追蹤成本是否超支
TIME_AND_MATERIAL	實報實銷，追蹤工時計費

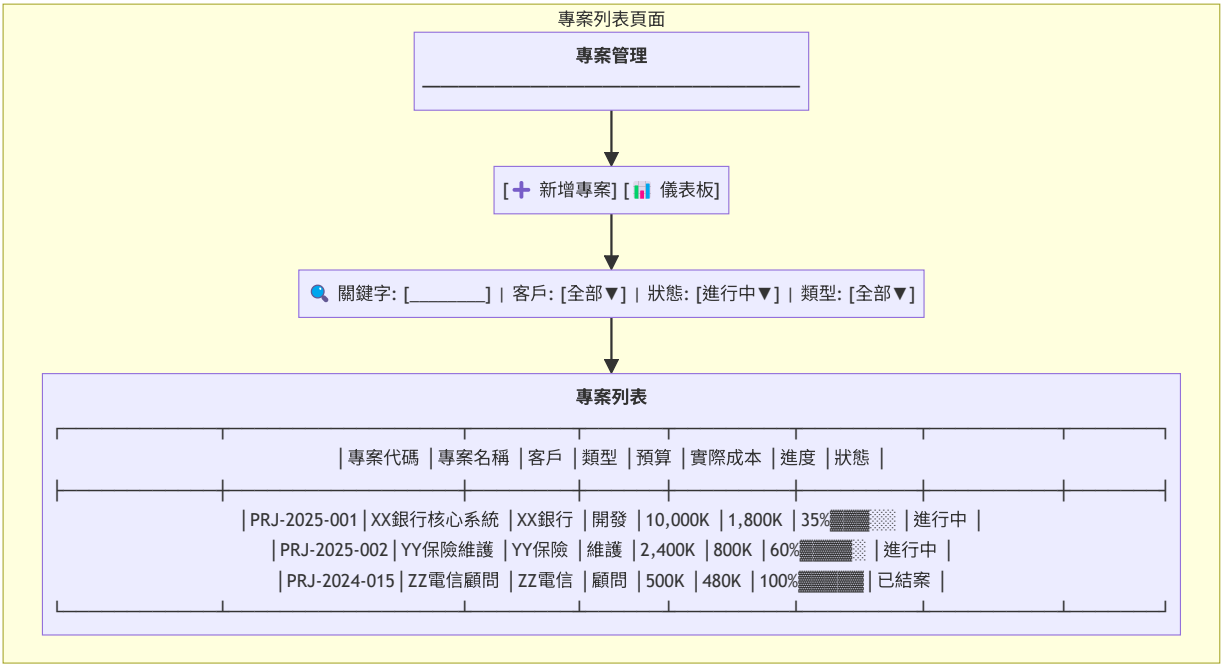
2. UI設計

2.1 頁面清單

頁面代碼	頁面名稱	路由	權限要求
`HR06-P01`	客戶管理頁面	`/admin/projects/customers`	customer:manage
`HR06-P02`	專案列表頁面	`/admin/projects`	project:read
`HR06-P03`	專案詳情頁面	`/admin/projects/:id`	project:read
`HR06-P04`	專案建立/編輯頁面	`/admin/projects/edit/:id?`	project:manage
`HR06-P05`	WBS工項管理頁面	`/admin/projects/:id/tasks`	project:task:manage
`HR06-P06`	專案成本分析頁面	`/admin/projects/:id/cost`	project:cost:read
`HR06-P07`	專案儀表板	`/admin/projects/dashboard`	project:read
`HR06-P08`	我參與的專案 (ESS)	`/profile/projects`	-
`HR06-M01`	客戶編輯對話框	(Modal)	customer:manage
`HR06-M02`	新增成員對話框	(Modal)	project:member:manage
`HR06-M03`	工項編輯對話框	(Modal)	project:task:manage

2.2 UI線稿 (Mermaid)

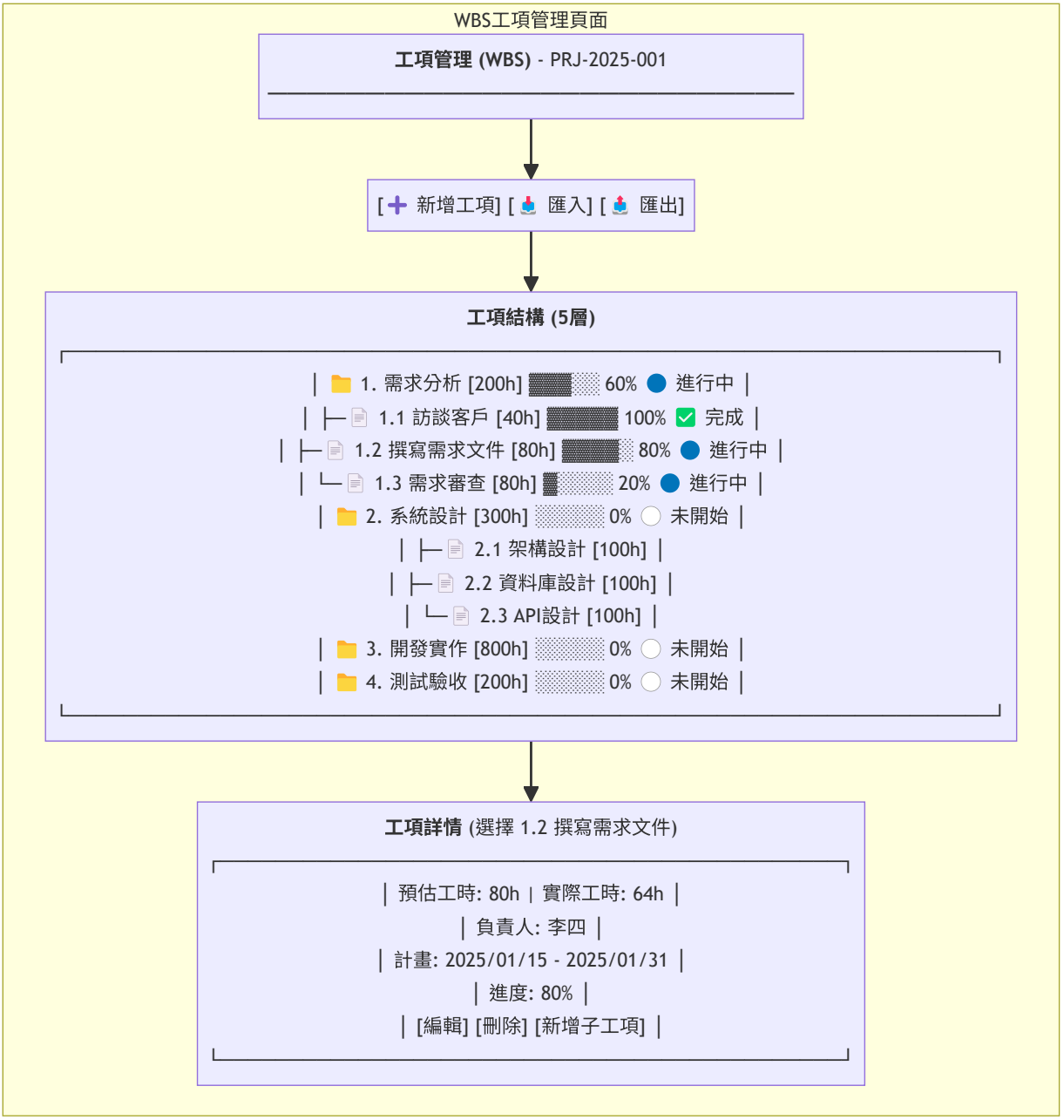
2.2.1 專案列表頁面 (HR06-P02)



2.2.2 專案詳情頁面 (HR06-P03)

```
┌─────────────────────────────────────────────────────────────────────────────┐
│                                 專案詳情頁面                                 │
│                                                                             │
│                   ┌─────────────────────────────────────────┐                │
│                   │   PRJ-2025-001: XX銀行核心系統開發       │                │
│                   │   ─────────────────────────────────────  │                │
│                   │                                         │                │
│                   │                                         │                │
│                   │                                         
```

2.2.3 WBS工項管理頁面 (HR06-P05)



2.2.4 專案成本分析頁面 (HR06-P06)

```
┌──────────────────────────────────────────────────────────────────────────────┐
│                                 專案成本分析頁面                                 │
│                                                                                  │
│                          專案成本分析 - PRJ-2025-001                             │
│                          ──────────────────────────                              │
│                                                                                  │
│                                      ↓                                           │
│                                                                                  │
│                                 成本摘要                                         │
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│                                                                                  │
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│                                      ↓                                           │
│                                                                                  │
│                                 成員成本分析                                     │
│                                                                                  │
│                                                                                  │
│                                                                                  │
│                                                                                  │
│                                                                                  │
│                                                                                  │
│                                                                                  │
│                                      ↓                                           │
│                                                                                  │
│                   📊 [燃盡圖] [成本曲線圖] [工時分布餅圖]                        │
│                                                                                  │
└──────────────────────────────────────────────────────────────────────────────┘
```
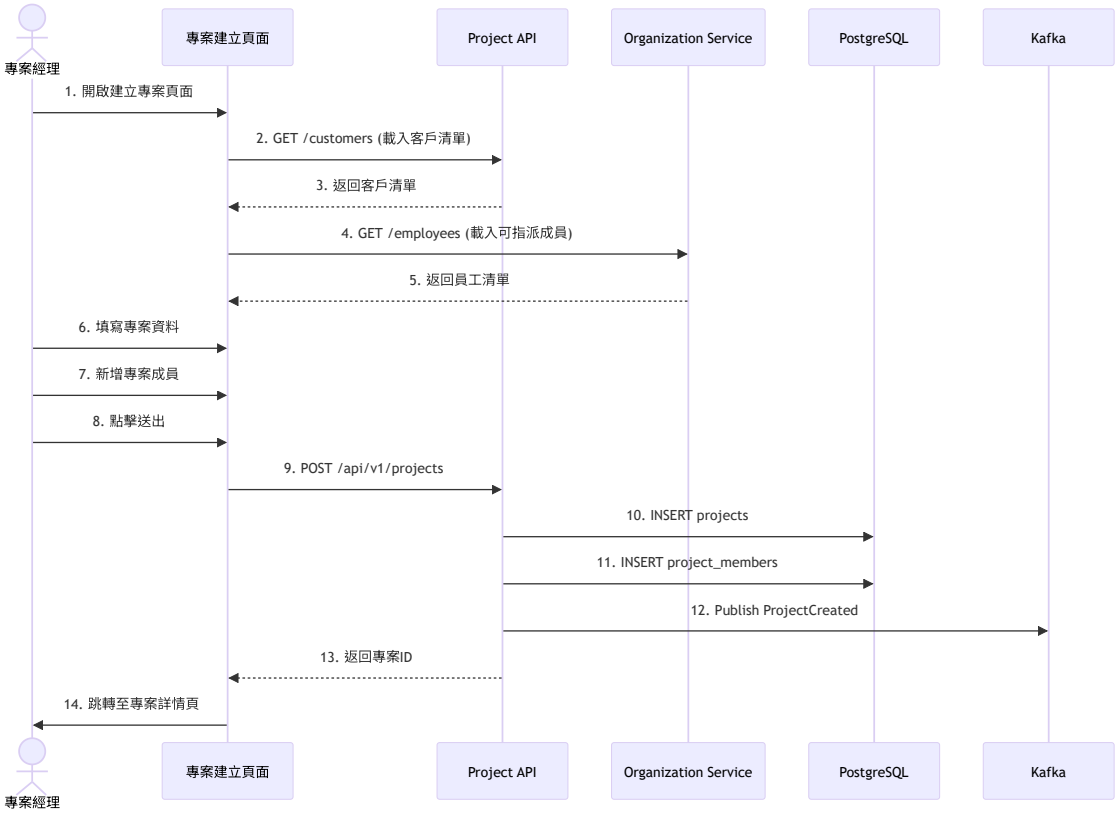
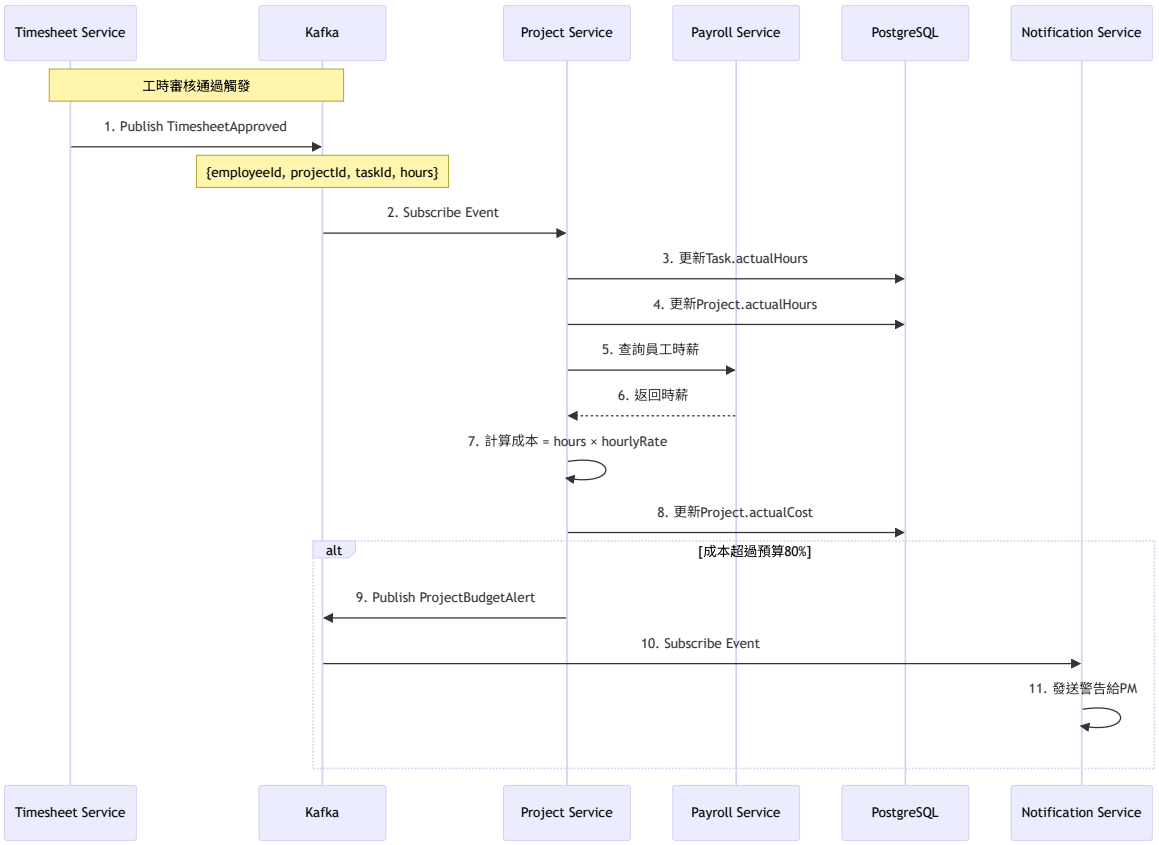

3. UX流程設計

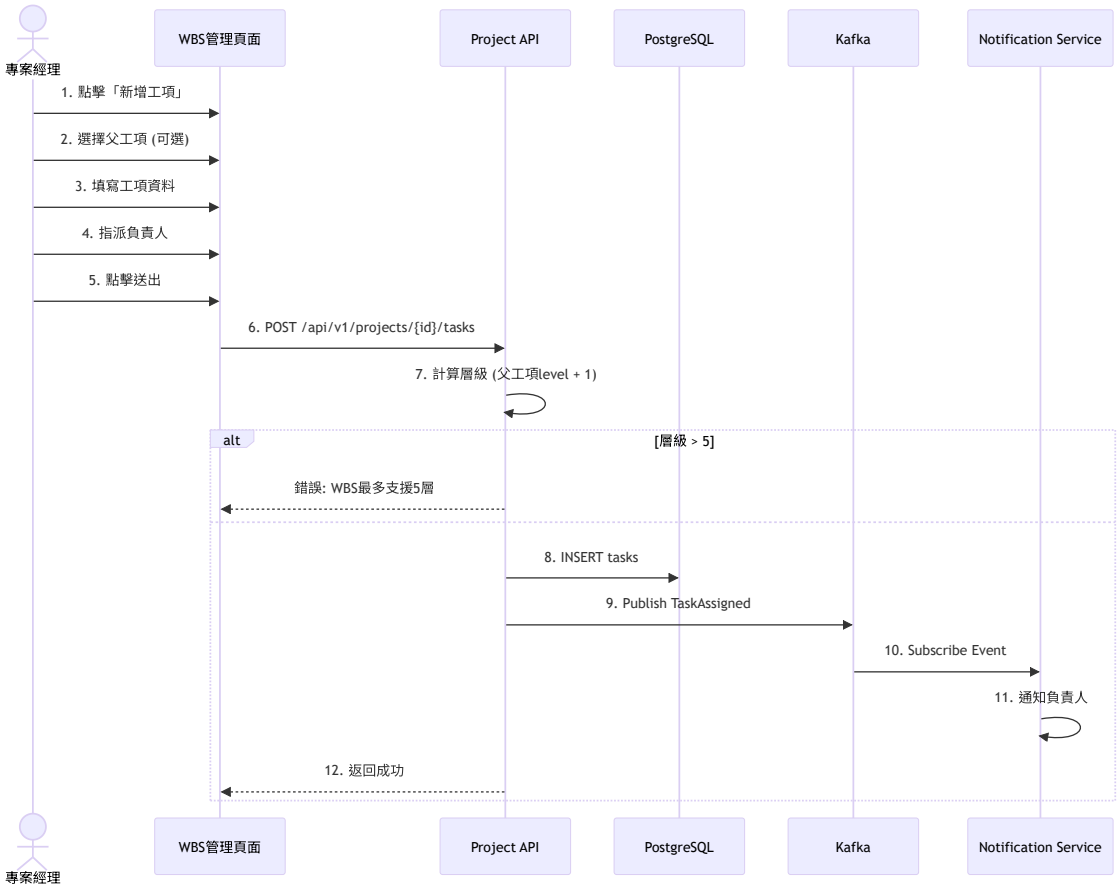
3.1 專案建立流程



3.2 專案成本更新流程 (Event-Driven)



3.3 WBS工項建立流程



4. 畫面事件說明

4.1 專案列表頁面事件 (HR06-P02)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-PRJ-01`	搜尋框	onChange	篩選專案	GET /api/v1/projects
`E-PRJ-02`	新增專案按鈕	onClick	跳轉建立頁面	-
`E-PRJ-03`	專案列點擊	onClick	跳轉專案詳情	-

4.2 WBS工項頁面事件 (HR06-P05)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-WBS-01`	新增工項按鈕	onClick	開啟新增工項對話框	-
`E-WBS-02`	工項節點點擊	onClick	顯示工項詳情	-
`E-WBS-03`	工項展開/收合	onClick	切換子工項顯示	-
`E-WBS-04`	確認新增工項	onClick	建立工項	POST /api/v1/projects/{id}/tasks
`E-WBS-05`	更新進度	onChange	更新工項進度	PUT /api/v1/tasks/{id}/progress
`E-WBS-06`	拖曳排序	onDrop	調整工項順序	PUT /api/v1/tasks/reorder

5. Data Flow設計

5.1 前端State結構

```
interface ProjectState {
  // 客戶
  customers: {
    list: Customer[];
    loading: boolean;
  };

  // 專案列表
  projects: {
    list: ProjectSummary[];
    filters: ProjectFilters;
    loading: boolean;
  };

  // 專案詳情
  currentProject: {
    data: ProjectDetail | null;
    members: ProjectMember[];
    loading: boolean;
  };

  // WBS工項
  tasks: {
    tree: TaskNode[];
    selectedTask: TaskDetail | null;
    loading: boolean;
  };

  // 成本分析
  costAnalysis: {
    summary: CostSummary | null;
    memberCosts: MemberCost[];
    loading: boolean;
  };

  // 我的專案 (ESS)
  myProjects: {
    list: MyProjectItem[];
    loading: boolean;
  };
}

interface TaskNode {
  taskId: string;
  taskCode: string;
  taskName: string;
```

```
  level: number;
  estimatedHours: number;
  actualHours: number;
  progress: number;
  status: TaskStatus;
  assignee: { id: string; name: string };
  children: TaskNode[];
}
```

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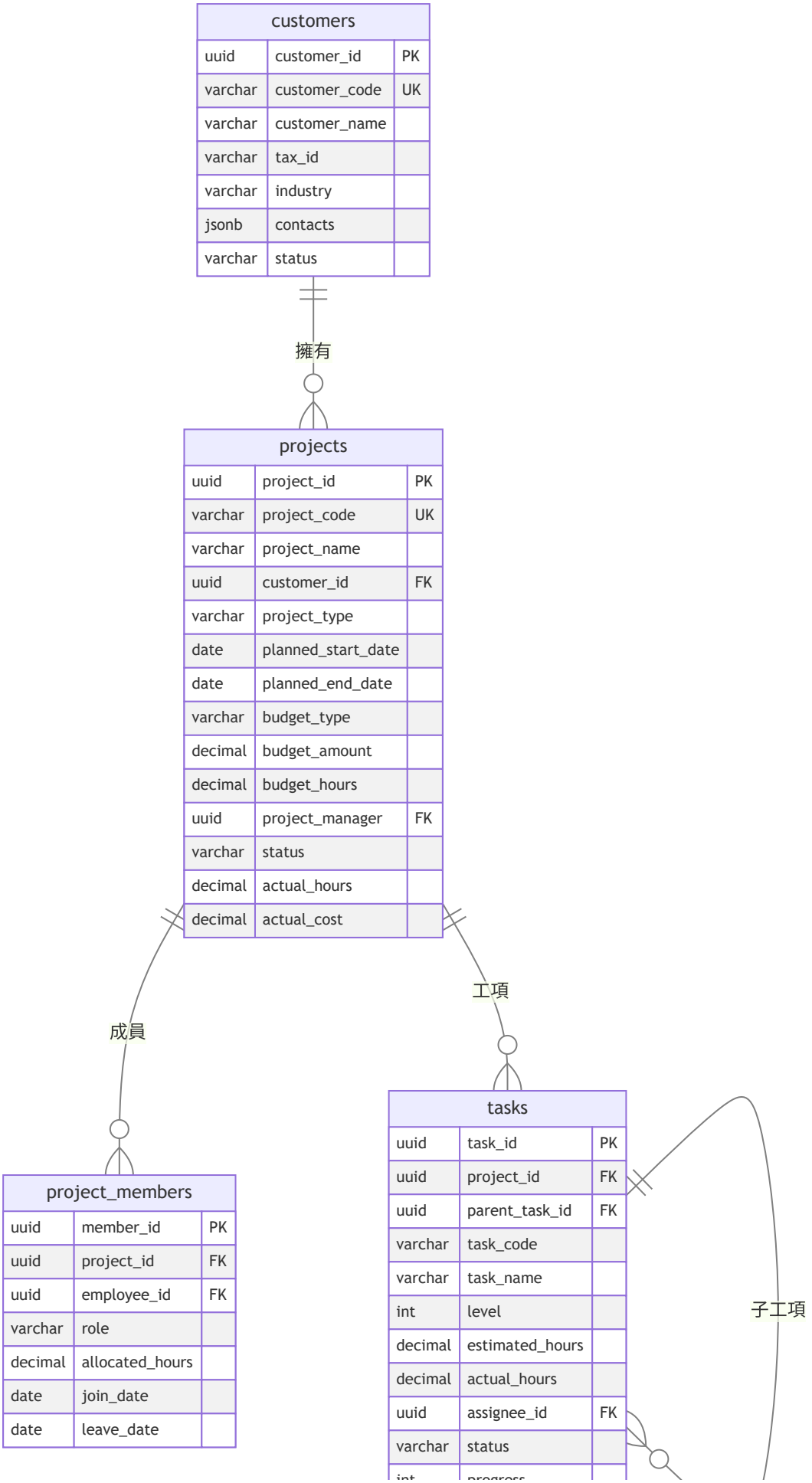
6. 資料庫設計

6.1 ER Diagram


customers
uuid customer_id PK
varchar customer_code UK
varchar customer_name
varchar tax_id
varchar industry
jsonb contacts
varchar status
擁有
projects
uuid project_id PK
varchar project_code UK
varchar project_name
uuid customer_id FK
varchar project_type
date planned_start_date
date planned_end_date
varchar budget_type
decimal budget_amount
decimal budget_hours
uuid project_manager FK
varchar status
decimal actual_hours
decimal actual_cost
成員
工項
project_members
uuid member_id PK
uuid project_id FK
uuid employee_id FK
varchar role
decimal allocated_hours
date join_date
date leave_date
tasks
uuid task_id PK
uuid project_id FK
uuid parent_task_id FK
varchar task_code
varchar task_name
int level
decimal estimated_hours
decimal actual_hours
uuid assignee_id FK
varchar status
子工項

int	progress	
int	display_order	

6.2 DDL Script

```
-- 客戶表
CREATE TABLE customers (
    customer_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    customer_code VARCHAR(50) NOT NULL UNIQUE,
    customer_name VARCHAR(255) NOT NULL,
    tax_id VARCHAR(20),
    industry VARCHAR(100),
    contacts JSONB DEFAULT '[]',
    address TEXT,
    phone_number VARCHAR(50),
    email VARCHAR(255),
    status VARCHAR(20) DEFAULT 'ACTIVE' CHECK (status IN ('ACTIVE', 'INACTIVE')),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_customer_name ON customers(customer_name);

-- 專案表
CREATE TABLE projects (
    project_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    project_code VARCHAR(50) NOT NULL UNIQUE,
    project_name VARCHAR(255) NOT NULL,
    customer_id UUID NOT NULL REFERENCES customers(customer_id),
    project_type VARCHAR(20) NOT NULL CHECK (project_type IN ('DEVELOPMENT', 'MAINTENANCE',
    planned_start_date DATE NOT NULL,
    planned_end_date DATE NOT NULL,
    actual_start_date DATE,
    actual_end_date DATE,
    budget_type VARCHAR(30) NOT NULL CHECK (budget_type IN ('FIXED_PRICE', 'TIME_AND_MATERI
    budget_amount DECIMAL(15,2),
    budget_hours DECIMAL(10,2),
    project_manager UUID NOT NULL,
    status VARCHAR(20) DEFAULT 'PLANNING'
        CHECK (status IN ('PLANNING', 'IN_PROGRESS', 'COMPLETED', 'ON_HOLD', 'CANCELLED')),
    actual_hours DECIMAL(10,2) DEFAULT 0,
    actual_cost DECIMAL(15,2) DEFAULT 0,
    description TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT chk_project_dates CHECK (planned_end_date >= planned_start_date)
);

CREATE INDEX idx_project_customer ON projects(customer_id);
CREATE INDEX idx_project_status ON projects(status);
CREATE INDEX idx_project_pm ON projects(project_manager);

-- 專案成員表
```

```sql
CREATE TABLE project_members (
    member_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    project_id UUID NOT NULL REFERENCES projects(project_id) ON DELETE CASCADE,
    employee_id UUID NOT NULL,
    role VARCHAR(100) NOT NULL,
    allocated_hours DECIMAL(10,2),
    hourly_rate DECIMAL(10,2),
    join_date DATE NOT NULL,
    leave_date DATE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT uk_project_member UNIQUE (project_id, employee_id)
);

CREATE INDEX idx_member_project ON project_members(project_id);
CREATE INDEX idx_member_employee ON project_members(employee_id);

-- 工項表 (WBS)
CREATE TABLE tasks (
    task_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    project_id UUID NOT NULL REFERENCES projects(project_id) ON DELETE CASCADE,
    parent_task_id UUID REFERENCES tasks(task_id),
    task_code VARCHAR(50) NOT NULL,
    task_name VARCHAR(255) NOT NULL,
    description TEXT,
    level INTEGER NOT NULL CHECK (level BETWEEN 1 AND 5),
    planned_start_date DATE,
    planned_end_date DATE,
    actual_start_date DATE,
    actual_end_date DATE,
    estimated_hours DECIMAL(10,2) NOT NULL DEFAULT 0,
    actual_hours DECIMAL(10,2) DEFAULT 0,
    assignee_id UUID,
    status VARCHAR(20) DEFAULT 'NOT_STARTED'
        CHECK (status IN ('NOT_STARTED', 'IN_PROGRESS', 'COMPLETED', 'BLOCKED')),
    progress INTEGER DEFAULT 0 CHECK (progress BETWEEN 0 AND 100),
    display_order INTEGER DEFAULT 0,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT uk_task_code UNIQUE (project_id, task_code)
);

CREATE INDEX idx_task_project ON tasks(project_id);
CREATE INDEX idx_task_parent ON tasks(parent_task_id);
CREATE INDEX idx_task_assignee ON tasks(assignee_id);

-- 專案成本快照表 (歷史記錄)
CREATE TABLE project_cost_snapshots (
    snapshot_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    project_id UUID NOT NULL REFERENCES projects(project_id),
```

```sql
    snapshot_date DATE NOT NULL,
    total_hours DECIMAL(10,2) NOT NULL,
    total_cost DECIMAL(15,2) NOT NULL,
    budget_utilization DECIMAL(5,2),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT uk_snapshot UNIQUE (project_id, snapshot_date)
);
```

---

7. Domain設計

7.1 聚合根

Project聚合根

```
@Entity
@Table(name = "projects")
public class Project {
    @EmbeddedId
    private ProjectId id;

    @Column(name = "project_code", unique = true, nullable = false)
    private String projectCode;

    @Column(name = "project_name", nullable = false)
    private String projectName;

    @Column(name = "customer_id", nullable = false)
    private UUID customerId;

    @Enumerated(EnumType.STRING)
    @Column(name = "project_type", nullable = false)
    private ProjectType projectType;

    @Embedded
    private ProjectSchedule schedule;

    @Embedded
    private ProjectBudget budget;

    @Column(name = "project_manager", nullable = false)
    private UUID projectManager;

    @OneToMany(cascade = CascadeType.ALL, orphanRemoval = true)
    @JoinColumn(name = "project_id")
    private List<ProjectMember> members = new ArrayList<>();

    @Enumerated(EnumType.STRING)
    @Column(name = "status", nullable = false)
    private ProjectStatus status;

    @Column(name = "actual_hours")
    private BigDecimal actualHours = BigDecimal.ZERO;

    @Column(name = "actual_cost")
    private BigDecimal actualCost = BigDecimal.ZERO;

    // ========== Factory Method ==========
```

```java
    public static Project create(CreateProjectCommand cmd) {
        Project project = new Project();
        project.id = ProjectId.generate();
        project.projectCode = cmd.getProjectCode();
        project.projectName = cmd.getProjectName();
        project.customerId = cmd.getCustomerId();
        project.projectType = cmd.getProjectType();
        project.schedule = new ProjectSchedule(
            cmd.getPlannedStartDate(),
            cmd.getPlannedEndDate()
        );
        project.budget = new ProjectBudget(
            cmd.getBudgetType(),
            cmd.getBudgetAmount(),
            cmd.getBudgetHours()
        );
        project.projectManager = cmd.getProjectManager();
        project.status = ProjectStatus.PLANNING;

        // 新增成員
        cmd.getMembers().forEach(m ->
            project.addMember(m.getEmployeeId(), m.getRole(), m.getAllocatedHours())
        );

        DomainEventPublisher.publish(new ProjectCreatedEvent(
            project.id.getValue(),
            project.projectCode,
            project.projectName
        ));

        return project;
    }

    // ========== Domain行為 ==========

    /**
     * 新增成員
     */
    public void addMember(UUID employeeId, String role, BigDecimal allocatedHours) {
        if (this.members.stream().anyMatch(m -> m.getEmployeeId().equals(employeeId))) {
            throw new DomainException("該成員已在專案中");
        }

        ProjectMember member = new ProjectMember(employeeId, role, allocatedHours);
        this.members.add(member);

        DomainEventPublisher.publish(new ProjectMemberAddedEvent(
            this.id.getValue(),
            employeeId,
            role
        ));
```

```java
    }

    /**
     * 開始專案
     */
    public void start() {
        if (this.status != ProjectStatus.PLANNING) {
            throw new DomainException("只有規劃中的專案可以開始");
        }
        this.status = ProjectStatus.IN_PROGRESS;
        this.schedule.setActualStartDate(LocalDate.now());
    }

    /**
     * 更新成本（被工時事件觸發）
     */
    public void updateCost(BigDecimal additionalHours, BigDecimal additionalCost) {
        this.actualHours = this.actualHours.add(additionalHours);
        this.actualCost = this.actualCost.add(additionalCost);

        // 檢查預算警告
        if (this.budget.getBudgetAmount() != null) {
            BigDecimal utilization = this.actualCost
                .divide(this.budget.getBudgetAmount(), 4, RoundingMode.HALF_UP)
                .multiply(new BigDecimal("100"));

            if (utilization.compareTo(new BigDecimal("80")) >= 0) {
                DomainEventPublisher.publish(new ProjectBudgetAlertEvent(
                    this.id.getValue(),
                    this.projectCode,
                    utilization
                ));
            }
        }
    }

    /**
     * 結案
     */
    public void complete() {
        if (this.status != ProjectStatus.IN_PROGRESS) {
            throw new DomainException("只有進行中的專案可以結案");
        }
        this.status = ProjectStatus.COMPLETED;
        this.schedule.setActualEndDate(LocalDate.now());

        DomainEventPublisher.publish(new ProjectCompletedEvent(
            this.id.getValue(),
            this.actualHours,
            this.actualCost
        ));
```

```java
    }

    /**
     * 計算進度
     */
    public int calculateProgress(List<Task> tasks) {
        if (tasks.isEmpty()) return 0;

        BigDecimal totalWeightedProgress = tasks.stream()
            .map(t -> t.getEstimatedHours().multiply(new BigDecimal(t.getProgress())))
            .reduce(BigDecimal.ZERO, BigDecimal::add);

        BigDecimal totalEstimatedHours = tasks.stream()
            .map(Task::getEstimatedHours)
            .reduce(BigDecimal.ZERO, BigDecimal::add);

        if (totalEstimatedHours.compareTo(BigDecimal.ZERO) == 0) return 0;

        return totalWeightedProgress
            .divide(totalEstimatedHours, 0, RoundingMode.HALF_UP)
            .intValue();
    }
}

// 值對象
@Embeddable
public class ProjectBudget {
    @Enumerated(EnumType.STRING)
    @Column(name = "budget_type")
    private BudgetType budgetType;

    @Column(name = "budget_amount")
    private BigDecimal budgetAmount;

    @Column(name = "budget_hours")
    private BigDecimal budgetHours;
}
```

Task聚合根 (WBS工項)

```java
@Entity
@Table(name = "tasks")
public class Task {
    @EmbeddedId
    private TaskId id;

    @Column(name = "project_id", nullable = false)
    private UUID projectId;

    @Column(name = "parent_task_id")
    private UUID parentTaskId;

    @Column(name = "task_code", nullable = false)
    private String taskCode;

    @Column(name = "task_name", nullable = false)
    private String taskName;

    @Column(name = "level", nullable = false)
    private int level;

    @Column(name = "estimated_hours")
    private BigDecimal estimatedHours;

    @Column(name = "actual_hours")
    private BigDecimal actualHours = BigDecimal.ZERO;

    @Column(name = "assignee_id")
    private UUID assigneeId;

    @Enumerated(EnumType.STRING)
    @Column(name = "status")
    private TaskStatus status;

    @Column(name = "progress")
    private int progress = 0;

    private static final int MAX_LEVEL = 5;

    /**
     * 建立工項
     */
    public static Task create(
            UUID projectId,
            UUID parentTaskId,
            int parentLevel,
            CreateTaskCommand cmd) {

        int newLevel = parentTaskId == null ? 1 : parentLevel + 1;
```

```java
        if (newLevel > MAX_LEVEL) {
            throw new DomainException("WBS最多支援" + MAX_LEVEL + "層");
        }

        Task task = new Task();
        task.id = TaskId.generate();
        task.projectId = projectId;
        task.parentTaskId = parentTaskId;
        task.taskCode = cmd.getTaskCode();
        task.taskName = cmd.getTaskName();
        task.level = newLevel;
        task.estimatedHours = cmd.getEstimatedHours();
        task.assigneeId = cmd.getAssigneeId();
        task.status = TaskStatus.NOT_STARTED;

        if (cmd.getAssigneeId() != null) {
            DomainEventPublisher.publish(new TaskAssignedEvent(
                task.id.getValue(),
                projectId,
                cmd.getAssigneeId(),
                task.taskName
            ));
        }

        return task;
    }

    /**
     * 更新進度
     */
    public void updateProgress(int newProgress) {
        if (newProgress < 0 || newProgress > 100) {
            throw new DomainException("進度必須在0-100之間");
        }

        this.progress = newProgress;

        if (newProgress == 100 && this.status != TaskStatus.COMPLETED) {
            this.status = TaskStatus.COMPLETED;
            DomainEventPublisher.publish(new TaskCompletedEvent(
                this.id.getValue(), this.projectId
            ));
        } else if (newProgress > 0 && this.status == TaskStatus.NOT_STARTED) {
            this.status = TaskStatus.IN_PROGRESS;
        }
    }

    /**
     * 累加實際工時
     */
```

```
    public void addActualHours(BigDecimal hours) {
        this.actualHours = this.actualHours.add(hours);
    }
}
```

---

8. 領域事件設計

事件名稱	觸發時機	發布服務	訂閱服務
`ProjectCreated`	建立專案	Project	-
`ProjectMemberAdded`	新增成員	Project	Timesheet
`ProjectCompleted`	專案結案	Project	Reporting
`ProjectBudgetAlert`	預算超過80%	Project	Notification
`TaskAssigned`	指派工項	Project	Notification
`TaskCompleted`	完成工項	Project	-

---

9. API設計

9.1 Controller命名對照

Controller	說明
`HR06CustomerCmdController`	客戶Command操作
`HR06CustomerQryController`	客戶Query操作
`HR06ProjectCmdController`	專案Command操作
`HR06ProjectQryController`	專案Query操作
`HR06TaskCmdController`	工項Command操作
`HR06TaskQryController`	工項Query操作
`HR06CostQryController`	成本分析Query操作

9.2 API總覽 (18個端點)

端點	方法	說明
`/api/v1/customers`	POST	建立客戶
`/api/v1/customers`	GET	查詢客戶列表
`/api/v1/customers/{id}`	GET	查詢客戶詳情
`/api/v1/projects`	POST	建立專案
`/api/v1/projects`	GET	查詢專案列表
`/api/v1/projects/{id}`	GET	查詢專案詳情
`/api/v1/projects/{id}`	PUT	更新專案
`/api/v1/projects/{id}/start`	PUT	開始專案
`/api/v1/projects/{id}/complete`	PUT	結案
`/api/v1/projects/{id}/members`	POST	新增成員
`/api/v1/projects/{id}/tasks`	POST	建立工項
`/api/v1/projects/{id}/tasks/tree`	GET	查詢WBS樹
`/api/v1/tasks/{id}`	PUT	更新工項
`/api/v1/tasks/{id}/progress`	PUT	更新進度
`/api/v1/projects/{id}/cost`	GET	查詢成本分析
`/api/v1/projects/my`	GET	我參與的專案 (ESS)

10. 工項清單摘要

前端開發工項

1. HR06-P01 客戶管理頁面
2. HR06-P02 專案列表頁面
3. HR06-P03 專案詳情頁面
4. HR06-P05 WBS工項管理頁面 (樹狀結構)
5. HR06-P06 專案成本分析頁面 (含圖表)
6. HR06-P08 我參與的專案 (ESS)

後端開發工項

1. Customer聚合根與Repository
2. Project聚合根與Repository
3. Task聚合根與Repository (樹狀查詢)
4. ProjectCostCalculationService
5. 客戶管理API (3端點)
6. 專案管理API (8端點)
7. 工項管理API (4端點)
8. 成本分析API (2端點)
9. 訂閱TimesheetApproved事件更新成本

資料庫開發工項

1. 建立5個資料表DDL
2. 建立索引
3. 初始化測試客戶與專案資料

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